POLYMER DENSITY / SAME-TEST-SET COMPARISON

v1 learned polymer density beyond classical group-contribution estimates

1,077 polymer structures from RadonPy PI1070; the model and physics baseline share the same 216-polymer test set.

Approach	How it was tested	MAE (g/cm3)	Fit (R2)
Van Krevelen physics	Random 60/20/20 train / calibrate / test	0.1327	0.026
Mean baseline	Same test polymers	0.1268	-
Amphora v1 / random forest	**646 train / 215 calibrate 216 test**	**0.0566**	**0.720**

57.3% less error

Learned molecular features improve density estimates over the named physics baseline.

Supports screening within the tested chemical distribution.

Source: RadonPy density comparison receipt [D1]. MD-simulated density at 300 K; random split.
Within-corpus result; novel-chemistry transfer is not established [A1].

v1 benchmarked thermal predictions against experimental polymer measurements

OpenPoly glass-transition temperature: a matched 89-polymer test set compares physics, learned chemistry and their combination.

Approach	How it was tested	MAE (K)	Fit (R2)
Transferred physics model	Van Krevelen Ridge; no OpenPoly refitting	80.30	-0.245
Physics + ML stack	Same train / calibrate / test partition	54.05	0.363
Amphora v1 / random forest	**265 train / 88 calibrate 89 test**	**48.77**	**0.405**

39.3% less error

On this matched test set, learned chemistry outperformed the transferred physics model.

A measured starting point for prioritizing thermal-property experiments.

Source: final physics / ML / stack comparison receipt [T1]. Experimental OpenPoly Tg; fresh random 60/20/20 split. Baseline was fitted on a different, simulated corpus. Single-corpus benchmark; not prospective lab validation [A1].

v1 improved solubility predictions for molecules withheld from training

BigSolDB: 94,532 measurement groups across 1,372 solutes, evaluated with leave-solute-out cross-validation.

Approach	How it was tested	RMSE (log S)
Best naive baseline	Solvent-mean baseline on the same evaluation	1.169
Data-cleaning stress test	Contradictory state-point groups removed; same protocol	0.874
Amphora v1	**Test solutes withheld from training**	**0.877**

25.0% less error

The advantage held after removing contradictory measurements: RMSE 0.877 became 0.874.

Relevant to solvent selection; this is not a drug-release benchmark.

Source: BigSolDB admissibility-hardened leave-solute-out audit [S1]. Full-corpus RMSE 95% CI: 0.842-0.915.
Literature-data benchmark and internal robustness audit; no prospective formulation or release claim.

v1 built an evidence trail behind the performance numbers

Named baselines, retained test protocols and explicit failure findings make the results inspectable.

01

Compare fairly

Density and thermal comparisons use matched test sets. Solubility holds entire solutes out of training.

02

Challenge the data

Removing contradictory solubility measurements preserved the gain: RMSE 0.877 to 0.874.

03

Keep the failures

Hybrid density improved RMSE, but not MAE. Novel-chemistry transfer failed; its limits remain explicit.

The investment case: an existing, inspectable foundation for choosing the next experiments.

Sources: comparison receipts [D1, T1], solubility stress test [S1], density stack [D2], external-validity audit [A1].
Retrospective computational evidence. Source receipts retain their original audit status; signatures were not re-verified for this deck.